
Chalmers University of Technology and Gothenburg University

Operating Systems
EDA093, DIT 401
Exam 2023-10-21

Date, Time: Wednesday 2024/01/03, 08.30-12.30

Course Responsible:

Vincenzo Gulisano (031 772 61 47)

Auxiliary material: You may have with you

- An English-Swedish, Swedish-English dictionary.
- No other books, notes, calculators, etc.

Grade-scale ("Betygsgränser"):

CTH: 3:a 30-39 p, 4:a 40-49 p, 5:a 50-60 p

GU: Godkänd 30-49p, Väl godkänd 50-60 p

Exam review ("Granskningstid"):

Will be announced after the exam.

Instructions

- Do not forget to write your personal number, if you are a GU or CTH student and at which program ("linje").
- Start answering each assignment on a new page; number the pages and use only one side of each sheet of paper.
- Write in a **clear manner** and **motivate** (explain, justify) your answers. If it is not clear what is written, your answer will be considered wrong. If it is not explained/justified, even a correct answer will get **significantly** lower (possibly zero) marking.
- Please make an effort to have nice calligraphy!
- If you make **any assumptions** in answering any item, do not forget to clearly state what you assume.
- Answer questions in English, if possible. If you have large difficulty with that and you think that your grade can be affected, feel free to write in Swedish.

Good luck !!!!

1. (4 p) Imagine that pure demand paging is used and that 3 frames are available and initially empty. What is the minimum number of page faults a process can experience?

[**HINT:** 1, because at least 1 page needs to be loaded. If more than 1 exists and, if it exists, whether it is used, we do not know. So the answer is 1.]

2. (4 p) Write the Effective Access Time formula using the following variables:

- p : the probability of a page fault
- m : memory access time (to access a given frame)
- o : page-fault serving overhead
- b : the probability of finding the dirty bit set to 1
- s_o : swap out time
- s_i : swap in time

assume there are no caches.

[**HINT:** $(1 - p) * m + p * (m + o + (1 - b) * s_o + s_i)$. $1 - p$ is the probability of no page fault, in that case, we simply pay m . In the rest of cases (p) we pay $m + o + s_o$ (but only $1 - b$) times, when the dirty bit is not set and we have to swap out, plus s_i]

3. (4 p) The documentation of the `vfork` system call says “`vfork()` is a special case of `clone(2)`. It is used to create new processes without copying the page tables of the parent process.” When is it a good and safe idea not to copy the page tables of the parent process?

[**HINT:** If the Child is immediately invoking `exec` and replacing its memory (and, as such, is own page table)]

4. (4 p) Suppose the pages maintained in the internal queue of the second-chance algorithm are 10 – 11 – 20 – 5 – 12 – 30, the associated bits, 1 – 0 – 1 – 0 – 0 – 1 and the next page that is going to be checked for replacement is 30. Which page will be chosen for replacement and what will the bits look like?

[**HINT:** if 1 it sets to 0. if 0 it selects it. The next that is checked is the last one (page 30), so that one and the first one (looping over) are set to 0 and the second one, which is already zero, is chosen. So 11 is chosen and the bits look like this 0 – 0 – 1 – 0 – 0 – 0]

5. (4 p) Suppose the following Access Control List is currently applied to handle accesses to file F1 for the users A, B, C, and D:

A: R, A: W, B: *, A: *, C: *, D: RW, D: R

Write an Access Control List that results in the same rights being applied to users but that is as compact as possible. Assume the 4 users are the only users that can exist in the system. You can use the wildcard * both for users as well as rights.

[**HINT:** A: R, D: RW, *: *, because Access Control Lists are scanned in order, and the first valid rule applies. Hence, any rule for user X preceded by another rule for the same user has no effect.]

6. (4 p) Explain why and by how much the n -bit salt mechanism weakens the effectiveness of dictionary attacks.

[**HINT:** for n bits, each word in the dictionary should be checked against 2^n variations, which makes it impractical for a large enough n]

7. (4 p) Write the pseudocode for the `testAndSet` instruction and show how it can be used to ensure mutual exclusion in the critical section problem.
[HINT: This is what is explained in slide 30 on the First synchronization lecture. Please check the slides]
8. (4 p) What are the four conditions that must hold simultaneously for a deadlock to occur?
[HINT: This is what is explained in slide 12 of the second synchronization lecture. Please check the slides.]
9. (8 p) Write Lamport's Bakery algorithm and explain why it represents a more decentralized version of Peterson's algorithm.
[HINT: this is what is explained in slide 31 of the second synchronization lecture. Please check the slides.]
10. (4 p) In which context is binary translation utilized? How does it work?
[HINT: In virtualization, binary translation is a virtualization mechanism in which the original binary code that belongs to e.g., a guest operating system is translated into a different binary code that gives greater control to the hypervisor/virtualizer, for instance, to define different behaviors for the guest software depending on whether interrupts are temporarily disabled or not]
11. (4 p) Describe in detail the difference between a guest-induced and a hypervisor-induced page fault.
[HINT: guest-induced means the guest OS itself is generating the page fault (because it misses a page in the respective frame, for instance). This fault, caught by the hypervisor, needs to be reinjected back to the guest OS. A Hypervisor induced page fault is generated in correspondence with an update of the shadow page table (maintained by the hypervisor itself)]
12. (4 p) What is the maximum serial portion a program can have if we want such a program to scale to at least 9X when using 10 cores instead of 1? Motivate your answer.
[HINT: Find S for Andhal's law for $1 / (S + (1-S)/10) = 9$. Any value smaller than or equal to that.]
13. (8 p) Describe how the lottery scheduling works. Make an example in which 4 single-threaded processes can be scheduled, P0, P1, P2, and P4. P0 has twice as many possibilities of being scheduled as P1, P1 has twice as many possibilities of being scheduled as P2, P2 has twice as many possibilities of being scheduled as P3, and P3 has twice as many possibilities of being scheduled as P4.
[HINT: Each process gets a certain number of "lottery tickets" (e.g., numbers). At each scheduling round, a ticket is selected randomly and the process that has such a ticket is scheduled for the next period/quantum. In the example, P4 1 ticket, P3 2 tickets, P2 4 tickets, and P1 8 tickets would work]